Evaluation of ML Models

BMED365: Topic List E01–E10

BMED365

Spring 2027

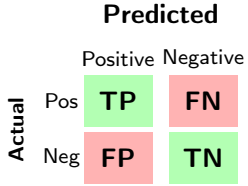
Overview

1. **Confusion Matrix**
   - E01–E02: TP, TN, FP, FN
2. **Fundamental metrics**
   - E03: Accuracy
   - E04: Precision
   - E05: Recall / Sensitivity
   - E06: Specificity
3. **Advanced metrics**
   - E07: F1-score
   - E08: ROC curve and AUC
   - E09: Imbalanced datasets
   - E10: TRIPOD guidelines

E01: Interpret a confusion matrix

What is it?

Table: predicted vs. actual classes



Reading the matrix:

- **TP**: Correct positive
- **TN**: Correct negative
- **FP**: False alarm
- **FN**: Missed case

Example: Cancer screening

- TP = Cancer correctly detected
- FN = Missed cancer (*dangerous!*)

E02: TP, TN, FP, FN in a medical context

Term	Definition	Medical example
TP	Sick patient, model says sick	Cancer correctly identified
TN	Healthy patient, model says healthy	Healthy person confirmed healthy
FP	Healthy patient, model says sick	Unnecessary biopsy
FN	Sick patient, model says healthy	Missed cancer

Critical question in medicine

What is worse: **False Positive** (FP) or **False Negative** (FN)?

- Screening: FN is more dangerous (missed disease)
- Invasive treatment: FP can be more dangerous (unnecessary risk)

E03: Accuracy

Definition

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

Example:

85	10
5	900

$$\text{Accuracy} = \frac{85 + 900}{85 + 900 + 10 + 5} = \frac{985}{1000} = 98.5\%$$

Warning

High accuracy can be misleading with imbalanced datasets!

E04: Precision (PPV)

Definition

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\text{True positives}}{\text{All predicted positives}}$$

Question: "Of all that the model predicts as positive, how many are actually positive?"

High precision is important when:

- Costly follow-up of positives
- Want to avoid false alarms
- Example: Rare disease

Example:

$$\text{Precision} = \frac{85}{85 + 5} = 94.4\%$$

"94% of flagged patients are actually sick"

E05: Recall / Sensitivity

Definition

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{\text{True positives}}{\text{All actual positives}}$$

Question: "Of all who are actually positive, how many does the model catch?"

High recall is important when:

- Severe consequence of missing cases (FN)
- Screening for dangerous disease
- Example: Cancer screening

Example:

$$\text{Recall} = \frac{85}{85 + 10} = 89.5\%$$

"We catch 89.5% of all cancer cases"

Other names

Recall = Sensitivity = True Positive Rate (TPR)

E06: Specificity

Definition

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{\text{True negatives}}{\text{All actual negatives}}$$

Question: "Of all who are actually negative, how many does the model correctly identify?"

High specificity is important when:

- Want to avoid unnecessary interventions
- Costly/risky treatment
- Example: Confirmatory tests

Example:

$$\text{Specificity} = \frac{900}{900 + 5} = 99.4\%$$

"99.4% of healthy correctly identified as healthy"

Trade-off

High sensitivity $\leftrightarrow$ Low specificity (and vice versa). Adjusting the threshold affects both.

E07: F1-score

Definition

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean of precision and recall

Why harmonic mean?

- Penalizes extreme values more harshly than arithmetic mean
- Precision = 100%, Recall = 0% $\rightarrow$ F1 = 0% (not 50%!)

When to use F1?

- Imbalanced datasets
- When both FP and FN are important
- Comparing models

Example:

$$F1 = 2 \cdot \frac{0.944 \cdot 0.895}{0.944 + 0.895} = 0.92$$

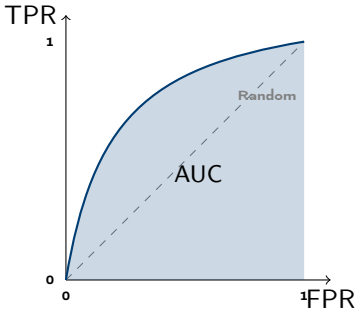
E08: ROC curve and AUC

ROC curve:

- **R**eceiver **O**perating **C**haracteristic
- Plots TPR vs. FPR for different thresholds
- Shows trade-off between sensitivity and specificity

AUC (Area Under Curve):

- Area under the ROC curve
- Value between 0 and 1
- AUC = 0.5: random guessing
- AUC = 1.0: perfect classification



Rule of thumb

AUC > 0.9: Excellent AUC 0.8–0.9: Good AUC 0.7–0.8: Fair AUC < 0.7: Poor

E09: When accuracy is insufficient

The problem with imbalanced datasets:

Example: 1000 patients

- 950 healthy, 50 sick
- Model predicts ALL as healthy

Results:

$$\text{Accuracy} = \frac{950}{1000} = 95\%$$

$$\text{Recall} = \frac{0}{50} = 0\%$$

High accuracy, but useless model!

Solutions:

- Use F1-score or AUC
- Focus on recall for screening
- Report all metrics
- Compare with baseline

Techniques for imbalance:

- Oversampling (SMOTE)
- Undersampling
- Class weights
- Stratified splitting

E10: TRIPOD Guidelines

What is TRIPOD?

Transparent **R**eporting of a multivariable prediction model for **I**ndividual **P**rognosis **O**r **D**iagnosis

Key points for reporting ML in medicine:

1. Clear description of study population
2. Define outcomes and predictors clearly
3. Describe missing data handling
4. Report model development in detail
5. Internal validation (cross-validation)
6. External validation if possible
7. Report calibration and discrimination
8. Discuss limitations

Why is it important?

TRIPOD ensures **reproducibility** and **quality control** of prediction models in health research.

Summary: E01–E10

Confusion Matrix

TP	FN
FP	TN

Metrics:

Recall $= \frac{TP}{TP+FN}$

Precision $= \frac{TP}{TP+FP}$

Specificity $= \frac{TN}{TN+FP}$

Advanced:

F1-score:
Combines Precision + Recall

ROC/AUC:
Overall picture independent of threshold

Key points

- Choose metric based on medical context (FN vs. FP consequences)
- Accuracy is often insufficient - use F1, AUC, precision, recall
- ROC/AUC gives overall picture independent of threshold
- Follow TRIPOD for transparent reporting